Business Requirement Document

E-Commerce Order Management Database System — Week 1

1. Introduction

This Business Requirement Document (BRD) defines the stakeholders, functional requirements, non-functional requirements, project scope, and assumptions/constraints for the proposed E-Commerce Order Management Database System, based on the business scenario and mandatory core entities (Customer, Category, Product, Supplier, Order, Order Details, Payment, Shipment, Review).

2. Stakeholder Analysis

Stakeholder	Role / Interest in the System
Customers	Register accounts, browse products, place orders, make payments, track shipments, and leave reviews.
Sales / Order Management Team	Process orders, monitor order status, and resolve order-related issues.
Inventory / Warehouse Staff	Maintain stock quantities, coordinate with suppliers, and prepare shipments.
Suppliers / Vendors	Supply products to the company and provide contact information for procurement.
Finance / Accounts Team	Record and reconcile payments, monitor payment status, and generate revenue reports.
Logistics / Shipment Team	Manage delivery addresses, dispatch shipments, and update delivery status.
Customer Support Team	Handle customer queries, complaints, and moderate/reply to product reviews.
Database Administrator (DBA)	Design, implement, secure, and maintain the underlying database system.
Business / Management Team	Use generated reports for decision-making, planning, and performance monitoring.

3. Functional Requirements

- The system shall allow new customers to register with Name, Email, Mobile Number, and Address.
- The system shall allow products to be created, updated, and assigned to a Category.
- The system shall track Stock Quantity for each product and reflect changes as orders are placed.
- The system shall allow suppliers and their contact information to be recorded and maintained.
- The system shall allow a customer to place an Order containing one or more Order Details (products, quantity, unit price).
- The system shall automatically compute the Total Amount of an Order from its Order Details.

- The system shall record Order Status (e.g., Pending, Confirmed, Shipped, Delivered, Cancelled) and allow it to be updated.
- The system shall record Payment details (Payment Method, Payment Date, Payment Status) for each Order.
- The system shall record Shipment details (Delivery Address, Shipment Date, Delivery Status) for each Order.
- The system shall allow customers to submit a Review (Rating and Comments) for products they have purchased.
- The system shall generate business reports such as sales by product, order history, stock levels, and customer activity.

4. Non-Functional Requirements

- **Performance:** The system shall retrieve order, product, and customer data within acceptable response times even as data volume grows.
- **Data Integrity:** The system shall enforce referential integrity between related entities (e.g., an Order Detail must reference a valid Order and Product) to prevent orphan or inconsistent records.
- **Data Redundancy Control:** The database shall be normalized to minimize duplicate storage of customer, product, and order information.
- **Security:** Customer personal and payment-related data shall be protected against unauthorized access.
- **Availability:** The database shall be available for order processing during business operating hours with minimal downtime.
- **Scalability:** The database design shall support growth in the number of customers, products, and orders without requiring structural redesign.
- **Usability:** The data structure shall be organized so that reports can be generated in a clear and meaningful format for business users.
- **Maintainability:** The database schema shall be documented and structured so that future modifications are straightforward.

5. Project Scope

In Scope:

- Customer registration and profile management.
- Product and category catalog management.
- Supplier information management.
- Order placement and order line-item (Order Details) management.
- Payment recording and status tracking.
- Shipment recording and delivery status tracking.
- Customer product reviews and ratings.
- Generation of business reports based on the stored data.

Out of Scope (for this phase):

- Front-end / user interface design and development.
- Third-party payment gateway integration logic.
- Real-time GPS-based shipment tracking.
- Marketing, promotions, and discount/coupon engines.
- Multi-currency and multi-language support.

6. Assumptions and Constraints

Assumptions:

- Each order belongs to exactly one registered customer.
- A product belongs to exactly one category.
- Every order eventually has one associated payment record and one associated shipment record.
- Customers can review only products they have purchased.
- Stock Quantity is updated only through recorded order transactions.

Constraints:

- Only the nine predefined core entities and their listed attributes may be used; no entities or attributes may be added, removed, or renamed.
- The database design must remain consistent with these entities throughout ER modeling, normalization, and SQL implementation.
- The project must be completed and documented within the semester project timeline.
- All deliverables must be submitted in the specified format via the GitHub repository.